

Hansini Rajesh

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Summary

AI and ML engineer with a computer science and applied statistics background, experienced in building end-to-end AI applications, integrating commercial LLMs into data pipelines, evaluating model reliability and profiling parallel systems for performance. Proficient in Python, C++ and PyTorch. MPS in Applied Statistics, Cornell University, May 2026.

Education

Cornell University

Ithaca, NY

Master of Professional Studies in Applied Statistics | GPA: 3.52/4.0 | Best MPS Capstone Project Award

Aug 2025 – May 2026

Coursework: Applied Neural Networks, Big Data Management, Probability Models & Inference, Multivariate Analysis.

Vellore Institute of Technology

Chennai, India

Bachelor of Technology in Computer Science | GPA: 3.82/4.0 (Scholaro-converted)

Sep 2021 – May 2025

Coursework: High Performance Computing, Natural Language Processing, Machine Learning, Data Structures & Algorithms.

Projects

Parallel N-Body Astrophysical Simulation

Jul 2024 – Nov 2024

Python, mpi4py, OpenMPI, NumPy, concurrent.futures

- Distributed $O(N^2)$ pairwise force computations across MPI ranks with hybrid MPI and thread-level parallelism, achieving 6.14x speedup on 8 ranks at 76.7% parallel efficiency.
- Benchmarked scaling with `MPI.Wtime()`, estimating an 8% serial fraction via Amdahl's Law; Allgather dominated overhead.

AI-Powered Indoor Navigation for the Visually Impaired

Jan 2025 – May 2025

Swift, iOS, ARKit, YOLOv8, A Pathfinding*

- Built an on-device AR navigation app combining ARKit scene understanding with a YOLOv8 object detector that flags indoor obstacles from the live camera stream and triggers spoken hazard alerts.
- Implemented A* pathfinding on vectorized floor plans for turn-by-turn routing, evaluated on success rate and path deviation.

LLM-Powered Earnings Signal Pipeline

Apr 2026

Python, Anthropic API, FinBERT, HuggingFace, statsmodels

- Built an end-to-end pipeline that ingests SEC EDGAR earnings call transcripts, automates feature extraction through the Anthropic API and scores sentiment with FinBERT, validated by event-study regression and a backtest.

Deep Learning for Multi-Frequency SZ Component Separation

2026

Python, PyTorch, Simulation

- Prototyped PyTorch architectures to separate Sunyaev-Zel'dovich signals from CMB and foregrounds, generating simulated sky maps as training data and benchmarking against classical ILC and matched filters.

CACTUS – Cornell Digital Agriculture Hackathon

2026

Python, PyTorch, CNN, MLP

- Built and shipped a dual-path CNN and MLP system fusing satellite imagery with soil sensor data, from ingestion to an interactive dashboard, within the hackathon deadline.

Experience

Research Fellow, Applied Statistics

Aug 2026 – Present

Cornell University

Ithaca, NY

- Building reproducible R analysis code and worked solutions for an eCornell course across approximately 15,000 event-level records.

Student Researcher, Applied Statistics

Jan 2026 – May 2026

Cornell University

Ithaca, NY

- Evaluated LLM reliability on structured coding across 460 variables and 585 documents against expert ground truth, measuring 71% agreement and building a taxonomy of model failure modes.
- Developed a Python NLP classification pipeline for approximately 2,000 free-text clinical records at Weill Cornell Medicine with automated multilingual translation.

Software Engineering Intern

Sep 2023 – Dec 2023

Punt Review

Dubai, UAE

- Built and tested responsive front-end components in HTML, CSS and JavaScript for an early-stage consumer platform.

Publications

- *Multi-Modal Driver Behavior Analysis and Speed Estimation using Fusion of Computer Vision and In-Car Sensor Data*, IEEE, 2024
- *LLM versus LL.M.: Comparing Human and AI Coding of Constitutional Law*, Ramdas et al., 2025 (Research Assistant)

Technical Skills

Programming: Python, C++, C, Java, JavaScript, Swift, SQL, R, MATLAB, HTML/CSS

AI & ML: PyTorch, TensorFlow, HuggingFace, Scikit-learn, XGBoost, YOLOv8, CNNs, NLP, LLM Evaluation, RAG, Anthropic API

Systems: MPI (mpi4py, OpenMPI), CUDA, Parallel Computing, Apache Spark, Hadoop, Linux, Git, Docker (basic), MySQL